MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 6423

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Triode
See figure 1
Mounting position -- Vertical, anode down
Weight -- 15 pounds nominal

ABSOLUTE RATINGS: F1 = 30 MHz

Parameter: Unit: Maximum:	Ef V	Eb kVdc	Ib Adc	Pi kW	Pp kkW	Ec kVdc	Ic Adc
B Audio:	7.0 +5%	12.5	2.5	30	12.5	----	---
B Telep:	7.0 +5%	12.5	1.8	19	12.5	----	---
C Telep: (Anode-modulated)	7.0 +5%	9.0	1.8	16	8.0	-1.4	0.5
C Teleg:	7.0 +5%	12.5	2.5	30	12.5	-1.4	0.5
TEST CONDITIONS:	7.0	17.5	---	--	----	----	---

ABSOLUTE RATINGS: F1 = 30 MHz

Parameter: Unit: Maximum:	Glass temp (See note 2) °C	Cooling (See note 1) ---
B Audio:	165	---
B Telep:	165	---
C Telep: (Anode-modulated)	165	---
C Teleg:	165	---
TEST CONDITIONS:	---	---

GENERAL

Qualification - Required (See note 4)

6423

(A) Denotes changes

FSC 5960

Source: https://assist.dla.mil -- Downloaded: 20261O-15T14:57Z
Check the source to verify that this is the current version before use.

PAR. NO.	TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS Min	LIMITS Max	UNIT
	Quality conformance inspection, part 1							
1201	Electrode current (grid)	Ec/Ib = 0.75 Adc; t = 300	0.65	II	Ic	----	-1.5	mAdc
1211	Electrode voltage (1) (anode)	Ec = -100 Vdc; Eb/Ib = 0.5 Adc	0.65	II	(1)Eb	14.5	18.0	kVdc
1261	Electrode voltage (2) (anode)	Ec = 0 Eb/Ib = 0.5 Adc	0.65	II	(2)Eb	6.5	8.5	kVdc
A 2214	RF useful output power	Eb = 17.5 kVdc; Ib = 2.5 Adc; Ic = 550 mAdc +20%; Rg = 2,500 ohms +-5%; F = 1 MHz +10%; t = 600	0.65	II	Po	27	---	kW
1231	Peak emission	eb = ec = 2,000 v	0.65	II	is	35	---	a
1261	Electrode voltage (grid)	Ec/Ib = 0.020 Adc	0.65	II	Ec	-190	-230	Vdc
1301	Filament current		0.65	II	If	80	90	Aac
	Quality conformance inspection, part 2							
1331	Direct-interelectrode capacitance	No shield	----	--	Cgp Cgk Cpk	20 25 --	28 35 2.5	pF pF pF
	Quality conformance inspection, part 3							
----	Service-life guarantee	See note 3	----	----	--	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:57Z
Check the source to verify that this is the current version before use.

NOTES:

1. Minimum airflow on anode cooler for 12.5 kW dissipation: 710 cfm at 4.7 inches of water (pressure) and 815 cfm at 5.1 inches of water (exhaust). Air distributor (see figure 2), must be used.

2. Environmental or operating conditions may make the use of an auxilliary airflow of 25-50 cfm necessary. This should be distributed to maintain uniform glass temperature, not greater than 165°C, around the circumference of the seals.

3. The tube manufacturer warrants the tube for 1 year, or 1,000 hours of heater life (whichever occurs first). This warranty applies only when the tube is operated within the maximum ratings. A defective tube shall either be replaced, or at the option of the contracting officer a credit shall be made in the amount of the original purchase price prorated on the basis of 1,000 hours of "heater-on" time.

(A) 4. Qualification. With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Product Lists is FSG 59 Engrg & Tec Spt Br, SMAMA (MMREE), Wright-Patterson AFB Oh 45433; however, information pertaining to qualification of products may be obtained from Defense Electronics Supply Center, Attention: DESC-EQA, Dayton OH 45420.

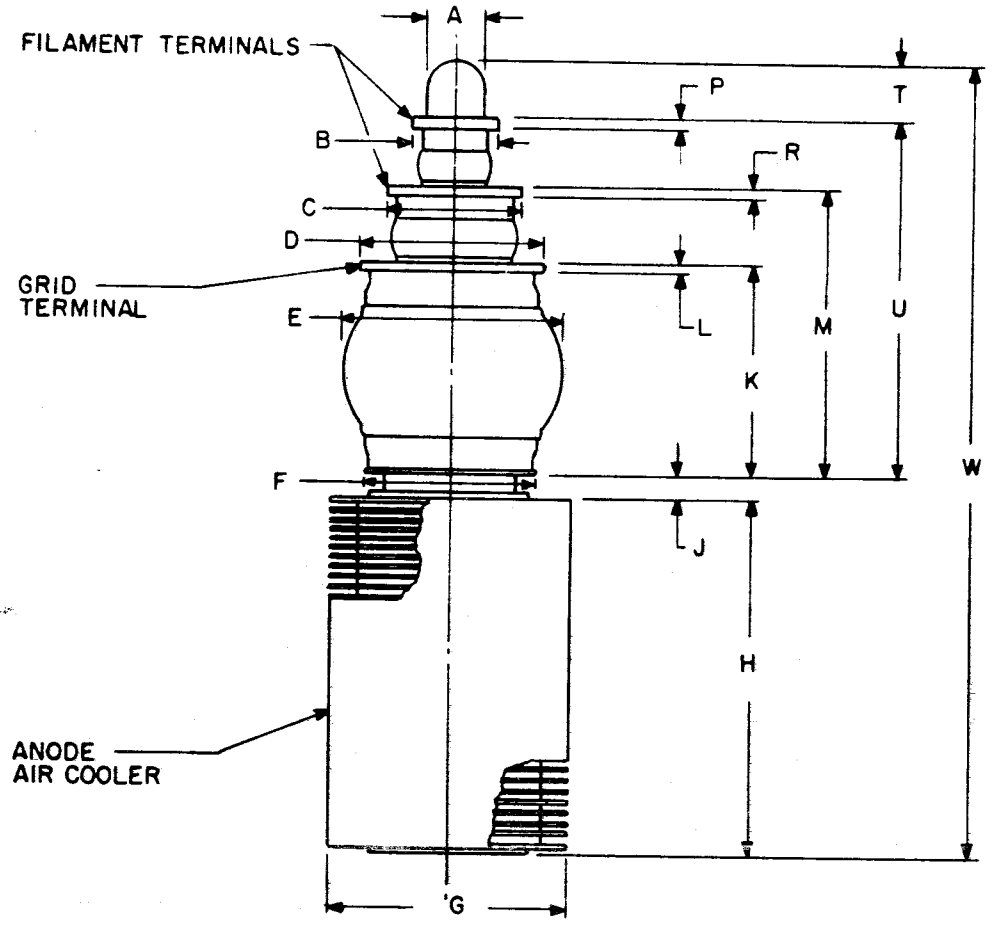
Custodian:
Air Force - 80

Preparing activity:
Air Force - 80

Reviewers:
Air Force - 11, 17,
DSA - ES

(Project 5960-F862)

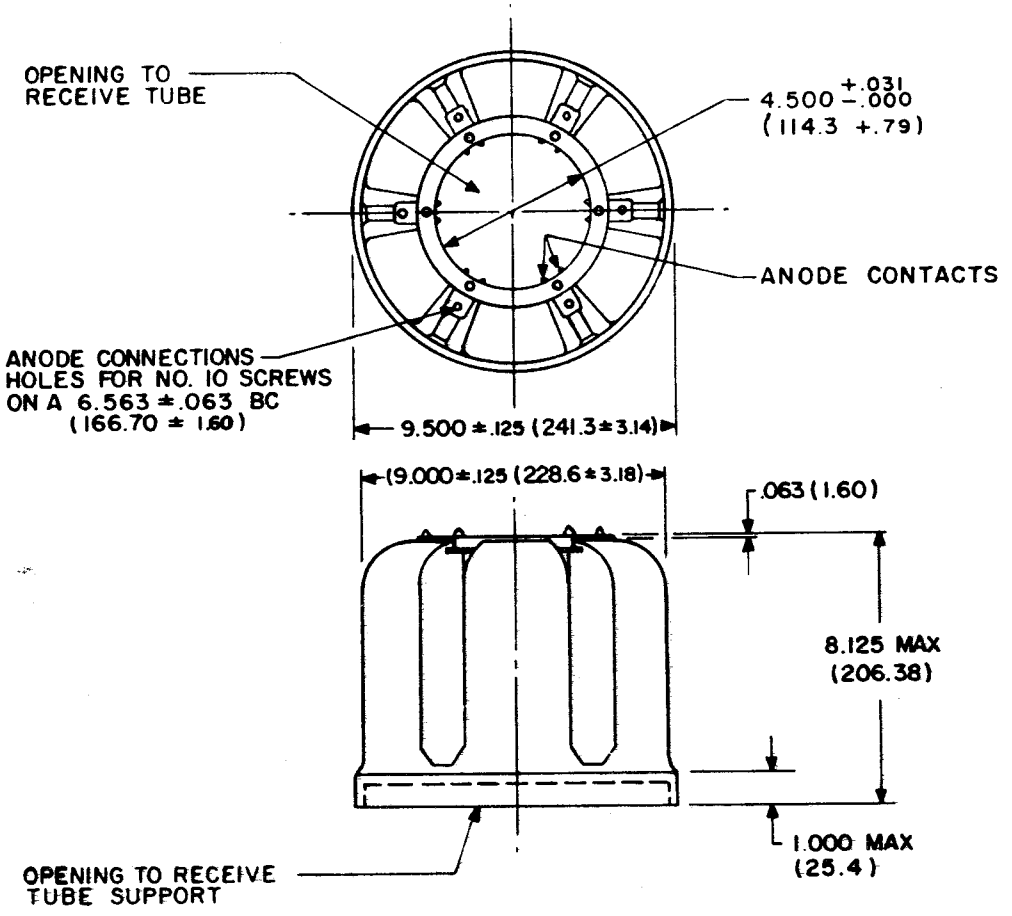
Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:57Z
Check the source to verify that this is the current version before use.



QUALITY CONFORMANCE INSPECTION, PART 2								
DIM	INCHES		MM		DIM	INCHES		MM	
	MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX
A	–	1.19	–	30.23	K	4.13	4.25	104.90	112.95
B	1.60	1.66	40.64	42.16	L	.16	.22	4.06	5.59
C	2.47	2.53	62.74	64.26	M	5.56	5.82	141.22	147.88
D	3.47	3.53	88.14	89.66	P	.16	.22	4.06	5.59
E	–	4.19	–	106.43	R	.16	.22	4.06	5.59
F	–	3.25	–	82.55	T	–	1.25	–	31.75
G	4.44	4.50	112.78	114.13	U	6.81	7.07	172.97	179.58
H	6.91	7.09	175.51	180.09	W	–	15.75	–	400.05
J	.19	–	4.83	–					

FIGURE 1. Outline Drawing of Electron Tube Type 5423.

Source: https://assist.dla.mil -- Downloaded: 20261O-15T14:57Z
Check the source to verify that this is the current version before use.



NOTE:

DIMENSIONS ARE IN INCHES. METRIC EQUIVALENTS IN PARENTHESES.

FIGURE 2. Outline Drawing of Air Distributor for Electron Tube Type 6427.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:57Z
Check the source to verify that this is the current version before use.